

Minimum order of a connected cubic bipartite graph with bondage number five

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Abstract

Every finite connected simple cubic bipartite graph on fewer than eighteen vertices has ordinary bondage number at most four. We prove this by an exhaustive exact computation whose completeness follows from a decomposition into three perfect matchings. The enumeration covers 35,759 connected constructions. Explicit vertex bijections reduce their verification to sixty reference graphs, each equipped with an edge-deletion witness of size at most four and exact domination certificates. Together with Yavari’s previously constructed eighteen-vertex graph of bondage number five, this determines the minimum order in the stated class. The accompanying certificates can be checked with the Python standard library, without trusting a graph generator or a canonical-labeling procedure.

1 Statement and relation to the known construction

All graphs are finite, simple and undirected. A set $D \subseteq V(G)$ is *dominating* if every vertex belongs to D or has a neighbor in D . Write $\gamma(G)$ for its minimum possible cardinality. The ordinary *bondage number* is

$$b(G) = \min\{|F| : F \subseteq E(G), \gamma(G - F) > \gamma(G)\}.$$

Here $G - F$ retains every vertex, including vertices isolated by the deletion; it need not be connected. Cubic means that every vertex has degree three.

Theorem 1. *Every connected cubic bipartite graph G with $|V(G)| < 18$ satisfies $b(G) \leq 4$. Consequently, eighteen is the minimum order of a connected cubic bipartite graph with bondage number five.*

The graph realizing the endpoint is due to Yavari [1, Theorem 1.2 and Sections 3–4], who gives its eighteen vertices, proves $\gamma = 6$, and establishes $b = 5$ by a five-edge witness and an exhaustive four-edge check. Our contribution is the complete exclusion of smaller graphs in this class. Rechecking the endpoint below corroborates the known construction. The inspected version of Yavari’s paper does not give this lower-order exclusion. A bounded literature search completed on 8 September 2026 found no equivalent exclusion; this is not a claim of exhaustive priority verification.

The proof separates a complete mathematical enumeration from the software used to discover useful certificates. This has broad methodological affinity with the certified reductions for regular graphs in Zeng [2]. That reference provides adjacent computational context, and no result from it is used to establish the present bondage bound.

2 An enumeration covering every smaller graph

If a cubic bipartite graph has parts A and B , edge counting gives $3|A| = 3|B|$. Write their common size as m . Simplicity implies $m \geq 3$. Thus the entire range below order eighteen is $m = 3, 4, 5, 6, 7, 8$.

Lemma 2. *Every cubic bipartite graph on $2m$ vertices is isomorphic to a graph on $\{a_0, \dots, a_{m-1}\} \cup \{b_0, \dots, b_{m-1}\}$ with edges*

$$\{a_i b_i, a_i b_{q(i)}, a_i b_{r(i)} : 0 \leq i < m\}, \quad (1)$$

where q consists of consecutive cycles of nondecreasing lengths at least two, and r ranges over all permutations satisfying

$$r(i) \notin \{i, q(i)\} \quad (0 \leq i < m). \quad (2)$$

Proof. A positive k -regular bipartite graph has a perfect matching. For $S \subseteq A$, counting the edges from S into $N(S)$ gives $k|S| \leq k|N(S)|$. To obtain a matching from this inequality, take a maximum matching and suppose some vertex $a \in A$ is unmatched. Explore alternating paths starting at a . No reached vertex in B is unmatched, since otherwise the matching could be augmented. Every reached vertex in B has a matched partner among the reached vertices in A , and the latter set additionally contains a . All neighbors of the reached A -vertices are reached, giving a set S with $|N(S)| < |S|$, a contradiction. The matching saturates A ; the equal part sizes make it perfect.

Apply this successively in degrees three and two. The remaining degree-one graph is itself a perfect matching. Hence the original edge set is a disjoint union $M_0 \cup M_1 \cup M_2$ of three perfect matchings. Label the partners in M_0 as a_i, b_i . The other matchings define permutations q, r . Their disjointness is precisely $q(i) \neq i$ and (2).

Relabeling both parts by the same permutation h replaces q, r by hqh^{-1}, hrh^{-1} and leaves the identity matching fixed. Permutations are conjugate exactly when they have the same cycle lengths. We may therefore put q into consecutive cycles, with lengths in nondecreasing order. Because q has no fixed point, all these lengths are at least two. Enumerating every admissible r includes the relabeled third matching. $\square$

For clarity, a partition $(\lambda_1, \dots, \lambda_t)$ of m with $2 \leq \lambda_1 \leq \dots \leq \lambda_t$ specifies cycles on the consecutive blocks of those lengths in $\{0, \dots, m-1\}$. The verifier recursively enumerates these partitions, loops over all $m!$ permutations r for each, and applies (2). A breadth-first traversal then discards disconnected graphs. Lemma 2 proves that this retains a representative of every connected graph required by Theorem 1. Repetitions are deliberate and harmless for a universal bound.

3 Domination certificates for the complete domain

Lemma 3. *Let H be a graph on n vertices and $F \subseteq E(H)$ with $1 \leq |F| \leq 4$. Suppose that integers $1 \leq g < h \leq n$ and sets $D, D' \subseteq V(H)$ satisfy: D dominates H and $|D| = g$; no $(g-1)$ -set dominates H ; D' dominates $H - F$ and $|D'| = h$; and no $(h-1)$ -set dominates $H - F$. Then $\gamma(H) = g$, $\gamma(H - F) = h$, and $b(H) \leq 4$. The same bondage bound holds for any graph isomorphic to H .*

Proof. A dominating set remains dominating when enlarged. Thus the absence of a dominating $(g-1)$ -set excludes every smaller size; the set D supplies the matching upper bound. Apply the same argument to $H - F$ and D' . The strict inequality $h > g$ proves the bondage bound. A vertex isomorphism transports domination, the edge set F , and the graph obtained by deletion. $\square$

The supplementary data contain sixty reference certificates. Each stores the full edge set of H , the integers g, h , the set F , and the upper-bound dominating sets. The verifier checks the graph hypotheses, the sizes and membership of the listed sets, and all $(g - 1)$ - and $(h - 1)$ -subsets required by Lemma 3. For a tested set S , the test is simply

$$\bigcup_{v \in S} (\{v\} \cup N_H(v)) = V(H). \quad (3)$$

The closed neighborhoods always include the vertex itself, including after edge deletion.

For every connected construction in (1), a saved record gives its cycle lengths, the permutation r , a reference index, and a map f of the full vertex set. The verifier checks that f is a bijection and that

$$\{\{f(u), f(v)\} : \{u, v\} \in E(G)\} = E(H).$$

It regenerates the complete enumeration rather than accepting a list of reported cases. The records must occur in the regenerated order, with no missing or extra row. Hence every retained construction is explicitly isomorphic to a reference satisfying Lemma 3.

Order	Cycle types	Permutations tested	Simple	Connected	References
6	1	6	1	1	1
8	2	48	6	6	1
10	2	240	25	25	2
12	4	2,880	322	321	5
14	4	20,160	2,317	2,311	13
16	7	282,240	33,192	33,095	38
Total	20	305,574	35,863	35,759	60

Table 1: Exact coverage of the three-matching enumeration. The columns “Simple” and “Connected” count constructions, with repetitions, rather than isomorphism classes. Pairwise nonisomorphism of the references is not needed or asserted.

The completed computation gives Table 1. All 35,759 bijections and all sixty reference certificates pass. The lower-bound tests in Lemma 3 examine 169,267 vertex subsets in total. These exact finite checks, combined with Lemma 2, prove the first assertion of Theorem 1.

4 The eighteen-vertex endpoint

For completeness we specify and recheck Yavari’s graph [1, Equations (3.1)–(3.2)]. Its vertex set is $\{0, 1, \dots, 17\}$. One bipartition is

$$X = \{0, 1, 4, 5, 7, 9, 13, 15, 17\}, \quad Y = \{2, 3, 6, 8, 10, 11, 12, 14, 16\}.$$

The following neighborhoods of the vertices in X specify all 27 edges:

x	$N(x)$	x	$N(x)$
0	$\{6, 10, 16\}$	9	$\{3, 10, 14\}$
1	$\{8, 11, 12\}$	13	$\{2, 6, 8\}$
4	$\{2, 3, 16\}$	15	$\{10, 11, 12\}$
5	$\{2, 3, 11\}$	17	$\{12, 14, 16\}$
7	$\{6, 8, 14\}$		

Proposition 4 (Yavari’s endpoint, exact replay). *The graph just specified is connected, cubic and bipartite, and has $\gamma(G) = 6$ and $b(G) = 5$.*

Proof. The graph hypotheses are checked directly from the edge list. Testing all $\binom{18}{5} = 8,568$ five-sets finds no dominating set. Testing all $\binom{18}{6} = 18,564$ six-sets gives exactly 297 dominating sets. Enlargement of dominating sets therefore gives $\gamma(G) = 6$.

As in Yavari [1, Equation (4.1)], delete

$$F_5 = \{\{0, 6\}, \{0, 10\}, \{0, 16\}, \{6, 7\}, \{6, 13\}\}.$$

No six-set dominates $G - F_5$, whereas $\{0, 1, 2, 4, 6, 10, 14\}$ does dominate it. Thus $\gamma(G - F_5) = 7$ and $b(G) \leq 5$.

For every one of the $\binom{27}{4} = 17,550$ four-edge subsets, the supplement supplies a six-vertex set dominating the remaining graph. Each witness is tested directly by (3). The number of surviving minimum dominating sets ranges from 1 to 103; in particular, none of these deletions increases γ . Deleting edges cannot decrease γ . If deleting fewer than four edges increased it, extending those edges to a four-set would still increase it. Hence $b(G) \geq 5$, proving the proposition. $\square$

Proposition 4 supplies the endpoint for the lower-order exclusion already proved, completing Theorem 1.

5 Reproduction and scope of verification

The accompanying archive `supplement.zip` includes the exact reference edge lists, deletion witnesses, all vertex maps, the endpoint graph and its four-edge survivor table, and verification programs. After extraction into a working copy, run these commands from its root:

```
python3 evidence/matching_census.py --verify
python3 evidence/verify_order18_cover.py
python3 verify_independently.py
```

They need only Python’s standard library and were checked with Python 3.9.6. The programs write replay reports in the extracted copy. The first command regenerates the complete smaller-order domain and verifies its certificates; the second verifies the endpoint and every four-edge survivor record.

The third command runs a verifier written in a separate mathematical review pass, which imports neither of the other implementations. It recognizes the canonical cycle permutations directly, generates the third matching by recursive placement of unused permissible images, and checks connectivity after contracting the identity matching. It checks every pair of vertices under each saved isomorphism. It also evaluates all vertex subsets in each reference graph and its deletion, and in the endpoint before and after F_5 : 5,976,704 subsets altogether. For the four-edge endpoint tests it checks, for every vertex outside a proposed dominating set, that its incident edge bundle to that set is not wholly deleted. All 35,759 constructions and 17,550 endpoint deletions passed this separate implementation.

Initial certificate discovery used `genbg` and `labelg` from nauty 2.9.3. Reference suggestions and vertex maps are subsequently checked by explicit edge-set equality; the mathematical completeness argument is Lemma 2. The decisive replay therefore does not depend on the correctness of a graph generator, a canonical-labeling algorithm, a SAT solver, or floating-point arithmetic. It remains a computer-assisted proof with ordinary interpreter and hardware trust, without proof-assistant formalization.

The result is confined to connected simple cubic bipartite graphs. It does not classify the exact bondage number of every smaller graph, establish uniqueness at order eighteen, or determine the minimum among nonbipartite or arbitrary graphs.

Computational and AI assistance. OpenAI Codex assisted with proof development, program and certificate preparation, literature searches, and manuscript drafting. A separate model context performed a mathematical review and wrote the additional verifier described above. This is a separate review pass, not external peer review or a claim of statistical independence. The mathematical evidence consists of the stated arguments and reproducible exact checks.

References

- [1] Yousof Yavari, *A Counterexample to Teschner's Bondage-Number Conjecture*, arXiv:2609.04257v1 (2026). <https://arxiv.org/abs/2609.04257>.
- [2] Zijian Zeng, *Proof-Carrying Reductions for 4-Regular Girth-Nine Graphs of Excess Four*, SSRN working paper (2026). <https://doi.org/10.2139/ssrn.7374858>.